



భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

Concentration Inequalities (AI2200)

Lecture 03

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Concentration of Bernoulli Averages

Theorem 1

Suppose $X_1, X_2, \dots \stackrel{\text{iid}}{\sim} \text{Ber}(p)$ for some fixed $p \in (0, 1)$. Then, the following hold.

- Upper tail probabilities:

$$\mathbb{P} \left(\left\{ \frac{S_n}{n} - p \geq \varepsilon \right\} \right) \begin{cases} = 0, & \varepsilon \geq 1 - p, \\ \exp(-n D_{\text{KL}}(\text{Ber}(p + \varepsilon) \parallel \text{Ber}(p))), & 0 < \varepsilon < 1 - p. \end{cases}$$

- Lower tail probabilities:

$$\mathbb{P} \left(\left\{ \frac{S_n}{n} - p \leq -\varepsilon \right\} \right) \begin{cases} = 0, & \varepsilon \geq p, \\ \exp(-n D_{\text{KL}}(\text{Ber}(1 - p + \varepsilon) \parallel \text{Ber}(1 - p))), & 0 < \varepsilon < p. \end{cases}$$

- Proof is left as **exercise**

Concentration of Bernoulli Averages

- Substituting $\varepsilon = \delta p$, we get that for all $\delta > 0$,

$$\mathbb{P} \left(\left\{ \frac{S_n}{n} - p \geq \delta p \right\} \right) \leq \exp \left(-n D_{\text{KL}}(\text{Ber}(p(1 + \delta)) \parallel \text{Ber}(p)) \right),$$

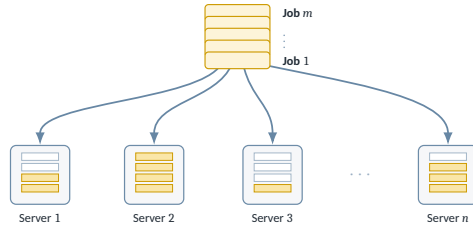
$$\mathbb{P} \left(\left\{ \frac{S_n}{n} - p \leq -\delta p \right\} \right) \leq \exp \left(-n D_{\text{KL}}(\text{Ber}(1 - p + \delta p) \parallel \text{Ber}(1 - p)) \right).$$

- After some algebra¹, it can be shown that for all $\delta > 0$,

$$\mathbb{P} \left(\left\{ \left| \frac{S_n}{n} - p \right| > \delta p \right\} \right) \leq 2 \exp \left(-\frac{n p \delta^2}{2 + \delta} \right).$$

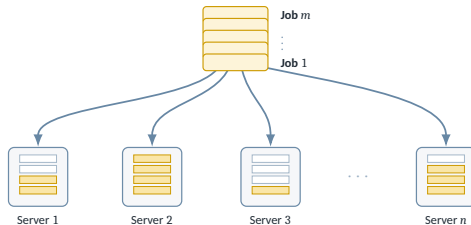
¹To be given as a guided exercise.

Application 1: Load Balancing



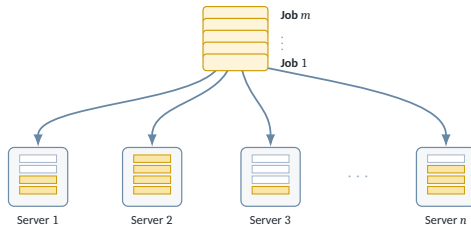
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- m jobs arrive sequentially and must be assigned to n servers with potentially different loads
- With no additional information, use uniform allocation:
send each arriving job to a uniformly random server
- Alternatively, send each job to the least-loaded server.
This requires all server loads, which are typically unavailable

Load Balancing under Uniform Allocation

- Define a collection of random variables $\{\chi_{ij} : i \in \{1, \dots, m\}, j \in \{1, \dots, n\}\}$ by

$$\chi_{ij} = \begin{cases} 1, & \text{job } i \text{ is allocated to server } j, \\ 0, & \text{otherwise.} \end{cases}$$

- For any fixed i , notice that

$$\mathbb{P}(\{\chi_{ij} = 1\}) = \frac{1}{n} = 1 - \mathbb{P}(\{\chi_{ij} = 0\}), \quad \chi_{ij} \sim \text{Ber}\left(\frac{1}{n}\right).$$

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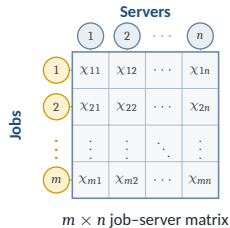
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- For any fixed server index j :

- The random variables

$$\chi_{1j}, \chi_{2j}, \dots, \chi_{mj} \stackrel{\text{iid}}{\sim} \text{Ber}\left(\frac{1}{n}\right).$$

- Load of server j is

$$L_j = \sum_{i=1}^m \chi_{ij}, \quad \mathbb{E}[L_j] = \frac{m}{n}.$$

$$\chi_{ij} \sim \text{Ber}\left(\frac{1}{n}\right).$$

		Servers			
		1	2	...	n
Jobs	1	χ_{11}	χ_{12}	...	χ_{1n}
	2	χ_{21}	χ_{22}	...	χ_{2n}
	...	...	...	...	...
	m	χ_{m1}	χ_{m2}	...	χ_{mn}

$m \times n$ job-server matrix

Load Balancing under Uniform Allocation

- Using the Bernoulli concentration from earlier, we have that for each $j \in \{1, \dots, n\}$ and $\delta > 0$,

$$\mathbb{P}\left(\left\{L_j - \frac{m}{n} \geq \frac{m\delta}{n}\right\}\right) \leq \mathbb{P}\left(\left\{\left|L_j - \frac{m}{n}\right| \geq \frac{m\delta}{n}\right\}\right) \leq 2 \exp\left(-\frac{(m/n)\delta^2}{2 + \delta}\right).$$

- As a consequence, we have that for any $\delta > 0$,

$$\begin{aligned} \mathbb{P}\left(\left\{\exists j \in \{1, \dots, n\} : L_j - \frac{m}{n} \geq \frac{m\delta}{n}\right\}\right) &= \mathbb{P}\left(\bigcup_{j=1}^n \left\{L_j - \frac{m}{n} \geq \frac{m\delta}{n}\right\}\right) \\ &\leq \sum_{j=1}^n 2 \exp\left(-\frac{(m/n)\delta^2}{2 + \delta}\right) && \text{(union bound)} \\ &\leq 2n \exp\left(-\frac{(m/n)\delta^2}{2 + \delta}\right). \end{aligned}$$

- In particular, if $m = n$, then

$$\mathbb{P}(\{\exists j \in \{1, \dots, n\} : L_j \geq 1 + \delta\}) \leq 2n \exp\left(-\frac{\delta^2}{2 + \delta}\right).$$

Load Balancing under Uniform Allocation

- If $\delta = \log(4n^4)$, then $\frac{\delta^2}{2 + \delta} \gg \frac{\delta}{2}$ for sufficiently large values of n , in which case

$$\mathbb{P}\left(\left\{\exists j \in \{1, \dots, n\} : L_j \geq 1 + \log(2n^2)\right\}\right) \leq \frac{1}{n}.$$

- Equivalently, we have

$$\mathbb{P}\left(\left\{\max_{j \in \{1, \dots, n\}} L_j \leq 1 + \log(2n^2)\right\}\right) \geq 1 - \frac{1}{n}.$$

Key Takeaway

If n incoming jobs are being allocated to n servers uniformly randomly, then the maximum load of any server is going to be at most $1 + O(\log n)$ with high probability.

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Application 2: Binary Classification

Which images contain a car?



Pic credits: Codex + GPT 5.6 Sol

Training data

Image	Label
X_1	$Y_1 = 1$
X_2	$Y_2 = 0$
X_3	$Y_3 = 1$
$\vdots$	$\vdots$
X_n	$Y_n = 1$

1 = car

0 = no car

$$(X_1, Y_1), \dots, (X_n, Y_n) \stackrel{\text{iid}}{\sim} P_{X,Y}$$

Application 2: Binary Classification

- We are given n labelled data points $(X_1, Y_1), \dots, (X_n, Y_n) \stackrel{\text{iid}}{\sim} P_{X,Y}$
- Suppose that $X_1, \dots, X_n$ take values in a set $\mathcal{X}$
Suppose that $Y_1, \dots, Y_n$ take values in a set $\mathcal{Y}$
- We want to train a classifier $g_n : \mathcal{X} \rightarrow \mathcal{Y}$ using the data $\mathcal{D}_n = \{(X_1, Y_1), \dots, (X_n, Y_n)\}$
- **Performance measure: probability of classification error (risk)**
Given a test data point $(X^*, Y^*) \sim P_{X,Y}$, we want g_n to have small risk

$P_{X,Y}$ unknown

$$R(g_n) := \mathbb{P}(\{g_n(X^*) \neq Y^*\}).$$

- Let $\mathcal{G}$ be the set of all classifiers based on n data points,

$$g_n^* := \arg \min_{g_n \in \mathcal{G}} \mathbb{P}(\{g_n(X^*) \neq Y^*\})$$

be the **optimal classifier** in $\mathcal{G}$

- For now, we will assume that $\mathcal{G}$ is a finite set

Empirical Risk Minimisation (ERM)

- Because we do not know $P_{X,Y}$, we cannot determine the optimal classifier g_n^*
- Given a classifier g_n , let its **empirical risk** be defined as

$$\widehat{R}(g_n) = \frac{1}{n} \sum_{k=1}^n \mathbf{1}_{\{g_n(X_k) \neq Y_k\}}.$$

- Let the **empirical risk minimizer** be defined as

$$\widehat{g}_n := \arg \min_{g_n \in \mathcal{G}} \widehat{R}(g_n).$$

True Risk Minimiser

$$g_n^* = \arg \min_{g_n \in \mathcal{G}} \mathbb{P}(\{g_n(X^*) \neq Y^*\}).$$

Empirical Risk Minimiser

$$\widehat{g}_n = \arg \min_{g_n \in \mathcal{G}} \frac{1}{n} \sum_{k=1}^n \mathbf{1}_{\{g_n(X_k) \neq Y_k\}}.$$

- We expect $\underbrace{\widehat{R}(\widehat{g}_n)}_{\text{optimal empirical risk}}$ to be close to $\underbrace{R(g_n^*)}_{\text{optimal risk}}$ for large n